CUSTOMER CHURN

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HIGHLIGHTS

- Business Understanding
- Data Understanding
- Data Cleaning
- Primary Results
- Modelling
- Recommendations

BUSINESS UNDERSTANDING

- SyriaTel is losing revenue as customers move to competitors.
  - This costs more than retaining an existing or gaining a new one.

- We want to identify customers likely tot churn so that:
  - We can target them with retention offers and incentives.

- Objective: To build a predictive classification model to identify customers likely to churn.

DATA UNDERSTANDING

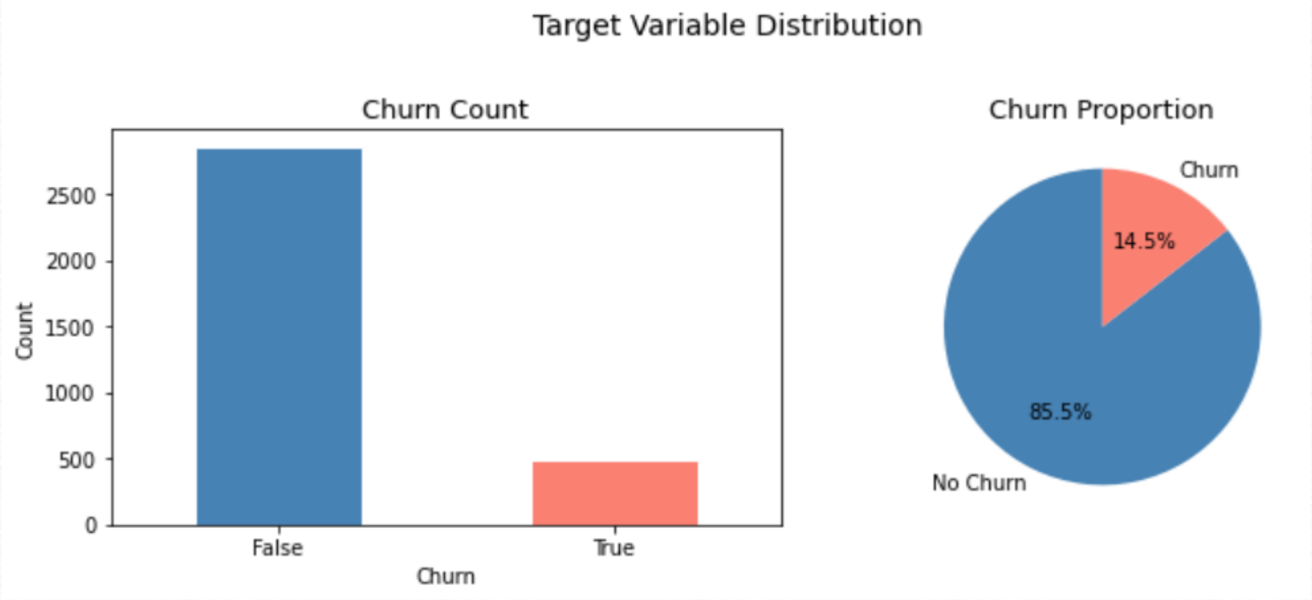
- Data sourced from:
  - SyriaTel Customer Churn Dataset from:
  - Link: https://www.kaggle.com/datasets/becksddf/churn-in-telecoms-dataset
  - Has 3,333 rows and 20 features
  - **Target variabl**e: Customer Churn (True/False)
  - F**eatures**: total day minutes, customer service calls, international plan, voice mail plan, total day charge

DATA CLEANING

- There were no:
  - Missing values.
  - Duplicated values.

PRIMARY RESULTS

- More customers did not churn as compared to those who did.

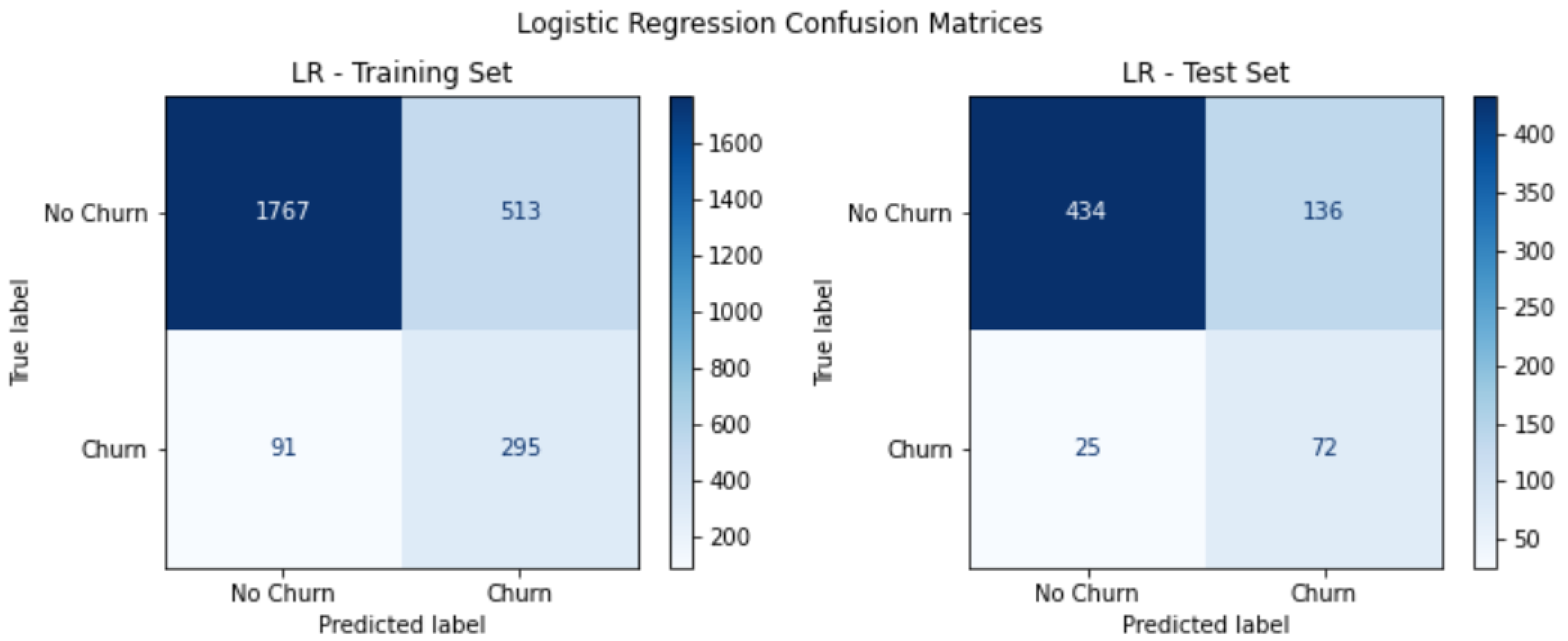


MODELLING

- For this project, we used:
  - Logistic Regression
  - Decision Tree
  - Random Forest Classifier.

- Metric for assessing success: Recall

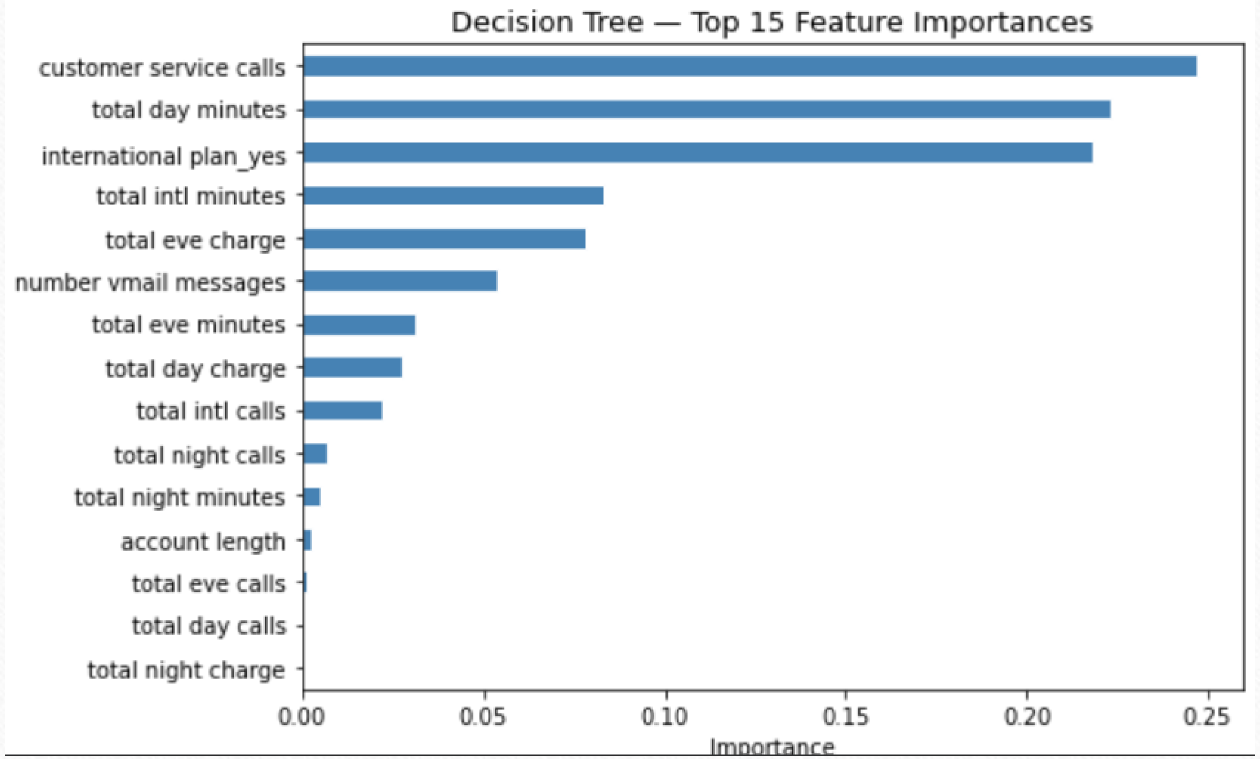
LOGISTIC REGRESSION



DECISION TREE

- **Test Recall: 0.722:** identifies 72% of actual churners on unseen data
- **Overfit gap: 0.133:** train recall (0.855) is much higher than test recall (0.722), showing the model memorizes training data more than Logistic Regression did
- **ROC-AUC: 0.803: S**lightly lower than Logistic Regression (0.828)

TOP FEATURES AS PER DECISION TREE

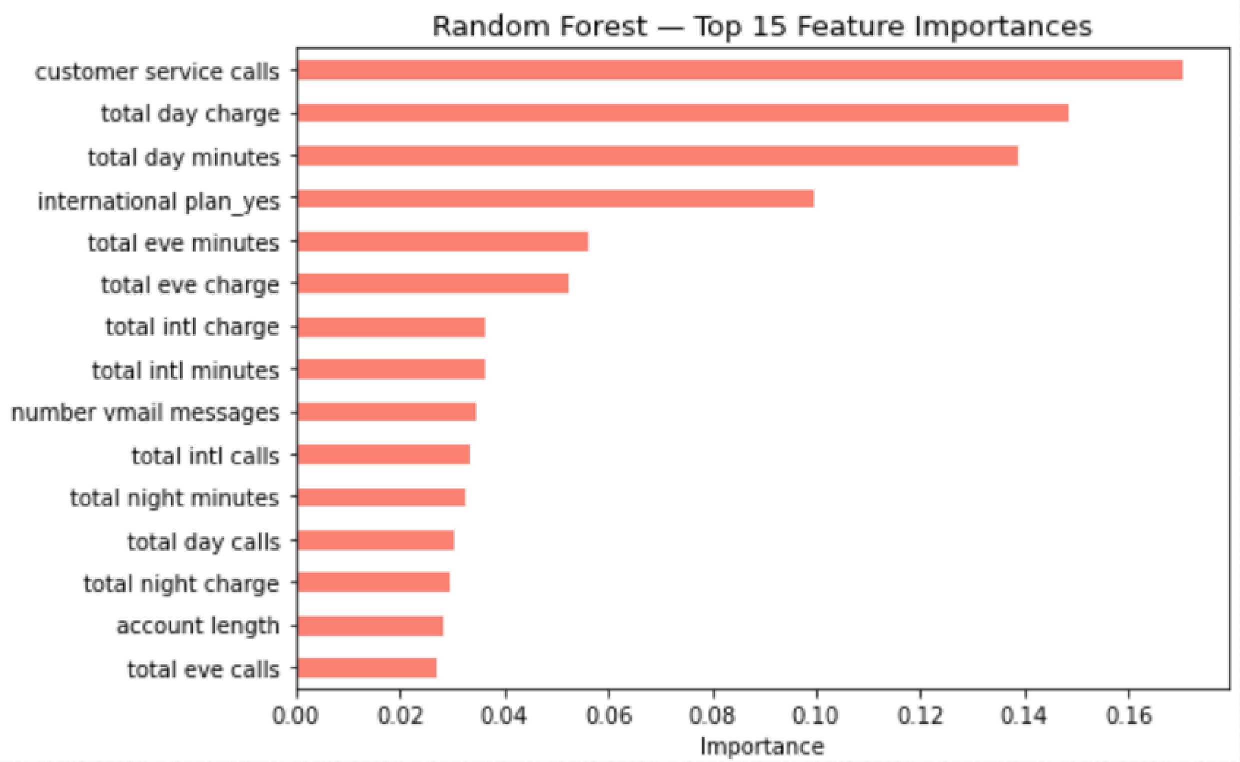


RANDOM FOREST CLASSIFIER

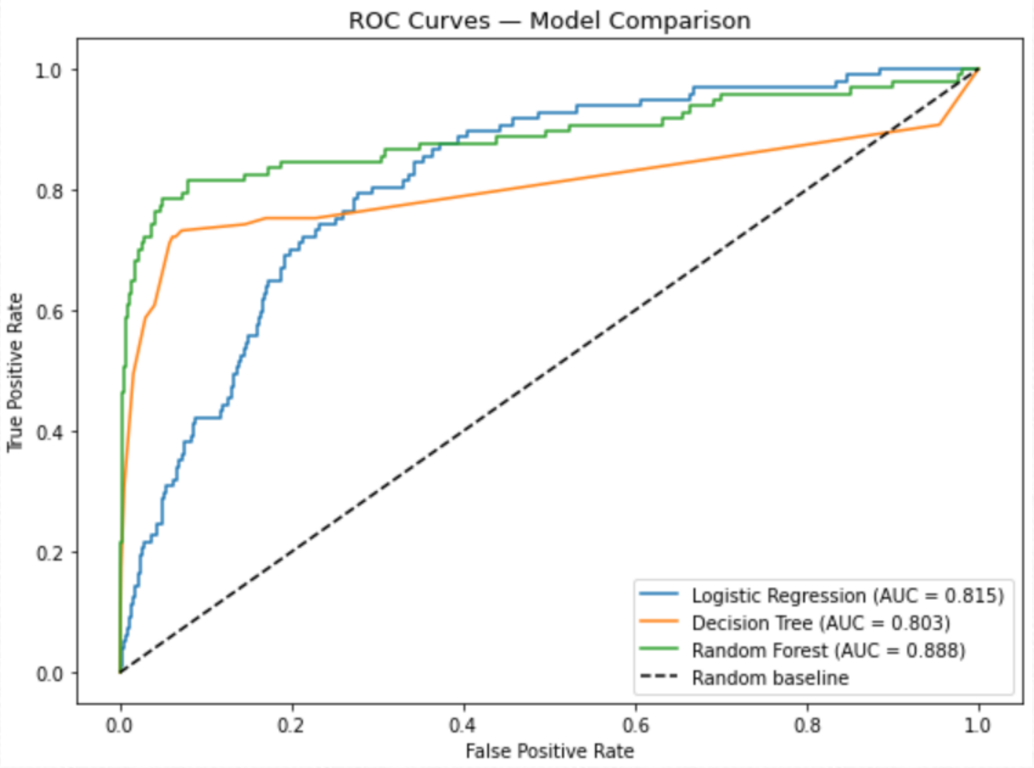
- Test recall (0.701) and has the largest overfit gap (0.208).

- Training ROC-AUC of 1.000:
    - Confirms it is memorizing training data rather than learning generalizable patterns.

TOP FEATURES AS PER RANDOM FOREST CLASSIFIER



ROC CURVE COMPARISON



CONCLUSION - MODEL

- **Logistic Regression** selected as the final model because it:
  - Achieved the **highest test recall (74.2%)**.
  - **Lowest overfit gap (0.022)**

RECOMMENDATIONS

- Deploy the model to score customers monthly.
- Intervene after the 2nd customer service call.
- Offer plan reviews to high day-charge customers.
- Investigate the international plan experience.


Thank you